

TuneStream (Music Streaming Service)

Description: Design a database for songs, artists, albums, playlists, and users.

Requirements:

- Manage music metadata and streaming rights.
- Track user playlists and listening history.
- Include subscription plans and payment info.

W3H:

What are functionalities do you want?	How you are going to achieve it?
1. Creating database for storing all tables. 2. Creating tables for storing the data efficiently 3. The tables should not have any redundancies 4. Each song must have unique identities	1. Creating a single Database to store all the application data. 2. Creating multiple tables to store the details of the application. 3. Creating separate tables for storing specific user ,songs details and connecting them by using foreign keys. 4. Creating a few numbers of tables only yo store all the application details. 5. The table data can have redundancy and duplicates 6. The table must have non-redundancy data 7. Each song may have unique identity as primary key. 8. Each song may have combined keys.
Why you have chosen this?	Why you have not chosen this?
1. Creating a single database to store all the information is the efficient way to handle the data. 2. Creating multiple tables to store the specific details helps to achieve normalization. 3. Each may have some unique identities like primary key.	1. Creating multiple databases may leads to high complexity of data handling. 2. Creating single table to store all the information may leads to redundancy of data. 3. Combined keys to each songs may harder to find the songs and handling it.

Database Name: Amplify

Tables included:

1. Songs
2. Artists
3. Albums
4. Playlists
5. Users

Creating a Database :

```
1 • create database Amplify;
```

Open a script file in this editor

Using Database:

```
2 • use Amplify;
```

Creating song Table:

```
create table Songs(songId int primary key, songName varchar(100) not null, artistId varchar(100) not null, duration varchar(20), movieName varchar(100));
```

Creating Artist Table:

```
create table Artists(artistId int primary key not null, artistName varchar(100) not null, industryName varchar(100));
```

Creating Albums Table:

```
create table Albums(albumId int primary key, albumName varchar(100), artistId int);
```

Creating Playlist table:

```
create table Playlists(playlistId int primary key not null , playlistName varchar(100) not null, songId int) ;
```

Creating user Table:

```
create table Users(userId int primary key auto_increment, userName varchar(100), subscrp varchar(100),  
payment varchar(100), favPlaylist varchar(100));
```

Adding foreign keys to establish the connection between the tables:

Songs and Albums:

```
Alter table Songs add constraint fk_artist1 foreign key Songs(songId) references Albums(albumId);
```

Songs and Artists:

```
Alter table Songs add constraint fk_artist foreign key Songs(songId) references Artists(artistId);
```

Albums and Artists:

```
Alter table Albums add constraint fk_albums foreign key(albumId) references Artists(artistId) ;
```

Playlists and Songs:

```
Alter table Playlists add constraint fk_play foreign key(playlistId) references Songs(songId) ;
```

User and Playlists:

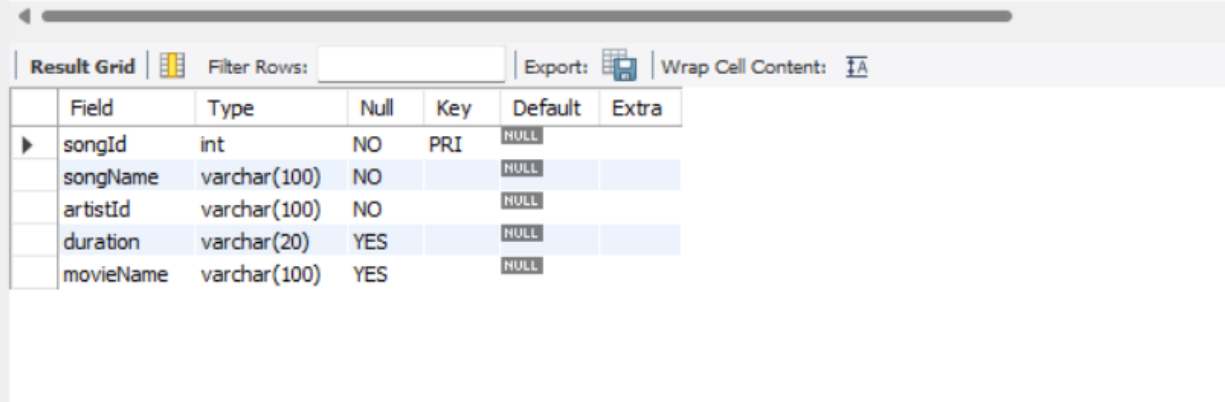
```
Alter table Users add constraint fk_user foreign key (userId) references Playlists(PlaylistId) ;
```

Table Structure of Songs Table:

26

27 • describe Songs;

28



	Field	Type	Null	Key	Default	Extra
▶	songId	int	NO	PRI	NULL	
	songName	varchar(100)	NO		NULL	
	artistId	varchar(100)	NO		NULL	
	duration	varchar(20)	YES		NULL	
	movieName	varchar(100)	YES		NULL	

Table Structure of Artists table:

28

29 • `describe Artists;`

30

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	artistId	int	NO	PRI	NULL	
	artistName	varchar(100)	NO		NULL	
	industryName	varchar(100)	YES		NULL	

Table Structure of Album table:

30

31 • `describe Albums;`

32

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	albumId	int	NO	PRI	NULL	
	albumName	varchar(100)	YES		NULL	
	artistId	int	YES		NULL	

Table Structure of Playlists table:

32

33 • `describe Playlists;`

34

Result Grid		Filter Rows:	Export:		Wrap Cell Content:	
	Field	Type	Null	Key	Default	Extra
▶	playlistId	int	NO	PRI	NULL	
	playlistName	varchar(100)	NO		NULL	
	songId	int	YES		NULL	

Table Structure of Users table:

Result Grid		Filter Rows:		Export:		Wrap Cell Content:	
	Field	Type	Null	Key	Default	Extra	
▶	userId	int	NO	PRI	NULL	auto_increment	
	userName	varchar(100)	YES		NULL		
	subscrp	varchar(100)	YES		NULL		
	payment	varchar(100)	YES		NULL		
	favPlaylist	varchar(100)	YES		NULL		

Inserting values into Artists Table:

```
• insert into Artists(artistId,artistName,industryName) values(101,'Gv Prakash','Kollywod'),  
(102,'Anirudh','Kollywood'),  
(103,'Ar Rahman','Bollywood');
```

Reading it:

Inserting values into Albums Table:

15

16 • `select * from Artists;`

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	artistId	artistName	industryName
▶	101	Gv Prakash	Kollywod
	102	Anirudh	Kollywood
	103	Ar Rahman	Bollywood
*	NULL	NULL	NULL

Table with Data:

26 • `select * from Albums;`

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Cor

	albumId	albumName
▶	111	Kathi
	121	Mersal
	123	Master
	124	Sarkar
	125	Mariyan

Inserting values with Playlists Table:

```
• insert into Playlists(playlistId,playlistName) values(201,'Love SOngs'),
(202,'MAss songs'),
(203,'Sad Songs'),
(204,'Motivational SOngs'),
(205,'Devotional SOngs');
```

Reading Data:

27

28 • **Select * from** Playlists;

29

30

Result Grid	Filter Rows:	Edit:	Export/Import:
playlistId	playlistName		
201	Love SOngs		
202	MAss songs		
203	Sad Songs		
204	Motivational SOngs		
205	Devotional SOngs		
NULL	NULL		

Playlists1 15

Inserting values into users Table:

2

```
3 • insert into Users(userId,userName,subscrp,payment,favPlaylist) values(11,'Barani','Standard','Gpay','Love Songs'),
4 (12,'Guna','Premium','Phonepe','MAss Songs'),
5 (13,'Kj','Standard','Creditcard','Love Songs'),
6 (14,'Deva','Standard','NetBanking','Devotional Songs'),
7 (15,'Rahul','Premium','Gpay','MAss Songs'),
8 (16,'Ra','Normal','Null','MAss Songs'),
9 (17,'Ramesh','Standard','BHIM','Love Songs'),
0 (18,'Mano','Normal','Null','Love Songs');
1
```

Data in Table:

53

54 • **select * from** Users1;

55

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Cor
userId	userName	subscrp	payment	favPlaylist
11	Barani	Standard	Gpay	Love Songs
12	Guna	Premium	Phonepe	MAss Songs
13	Kj	Standard	Creditcard	Love Songs
14	Deva	Standard	NetBanking	Devotional Songs
15	Rahul	Premium	Gpay	MAss Songs
16	Ra	Normal	Null	MAss Songs
17	Ramesh	Standard	BHIM	Love Songs
18	Mano	Normal	Null	Love Songs
NULL	NULL	NULL	NULL	NULL

COMPLETE CRUD OPERATION WITH SONGS TABLE:

Inserting values into Songs Table:

```
insert into Songs(songId,songName,duration,movieName) Values(1,'Para Para','5mins','Neer Paravai'),
(2,'Adiye','5mins','Bachelor'),
(3,'Kadal Rasa','5mins','Mariyan'),
(4,'Alaporan Tamizhan','5mins','Mersal'),
(5,'Neethane','5mins','Mersal'),
(6,'Vaathi Coming','5mins','Master'),
(7,'Kathi the rise','5mins','Kathi'),
(8,'Jananayagan the rise','5mins','JanaNayakan'),
(9,'Para Para-Remastered','5mins','Neer Paravai'),
(10,'Simtangan','5mins','Sarkar');
```

Reading Data from Songs Table:

50 • select * from Songs;

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
songId	songName	duration	movieName	albumId
1	Para Para	5mins	Neer Paravai	NULL
2	Adiye	5mins	Bachelor	NULL
3	Kadal Rasa	5mins	Mariyan	NULL
4	Alaporan Tamizhan	5mins	Mersal	NULL
5	Neethane	5mins	Mersal	NULL
6	Vaathi Coming	5mins	Master	NULL
7	Kathi the rise	5mins	Kathi	NULL
8	Jananayagan the rise	5mins	JanaNayakan	NULL
9	Para Para-Remastered	5mins	Neer Paravai	NULL
10	Simtangan	5mins	Sarkar	NULL
*	NULL	NULL	NULL	NULL

Updating the Songs Data:

```
Update Songs set songname='Aathi',duration='3mins' where songId=7 ;
```

```

53
54 • select * from Songs where SongId=7;
55

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

	songId	songName	duration	movieName	albumId
▶	7	Aathi	3mins	Kathi	NULL
•	NULL	NULL	NULL	NULL	NULL

Deleting the row in the table:

```

6 • Delete from Songs where SongId=7;

```

Data in Table after deleting the row with id=7;

```

57 • select * from Songs1;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

	songId	songName	duration	movieName	albumId
▶	1	Para Para	5mins	Neer Paravai	NULL
	2	Adiye	5mins	Bachelor	NULL
	3	Kadal Rasa	5mins	Mariyan	NULL
	4	Alaporan Tamizhan	5mins	Mersal	NULL
	5	Neethane	5mins	Mersal	NULL
	6	Vaathi Coming	5mins	Master	NULL
	8	Jananayagan the rise	5mins	JanaNayakan	NULL
	9	Para Para-Remastered	5mins	Neer Paravai	NULL
	10	Simtangan	5mins	Sarkar	NULL
•	NULL	NULL	NULL	NULL	NULL

Drop Table:

55

56 • `Drop table Users1;`

57

Sample ER Diagram:

